

AI Pricing in the Macroeconomy

Predictions for the Phillips Curve and Monetary Policy

KC FRB Brown Bag, September 16 2026

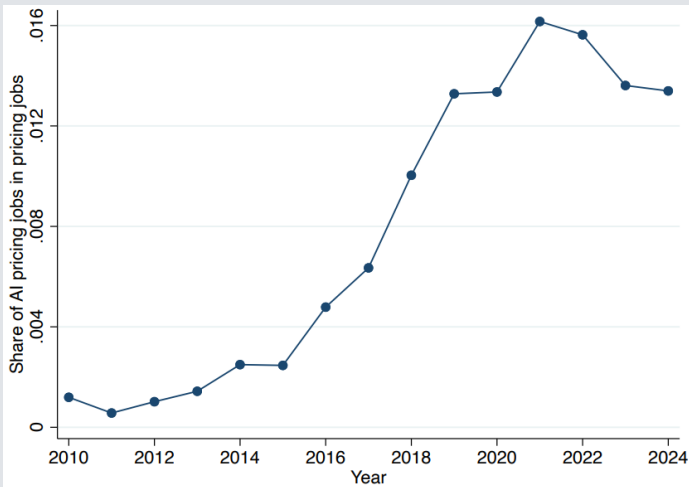
Jonathan J. Adams

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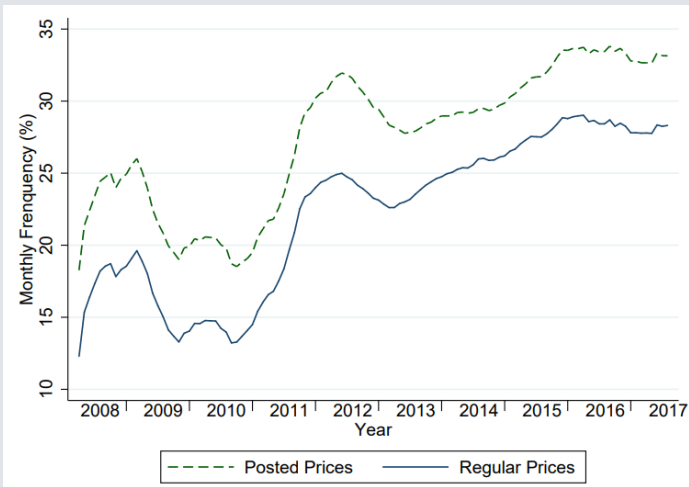
Motivation

Some Motivating Evidence



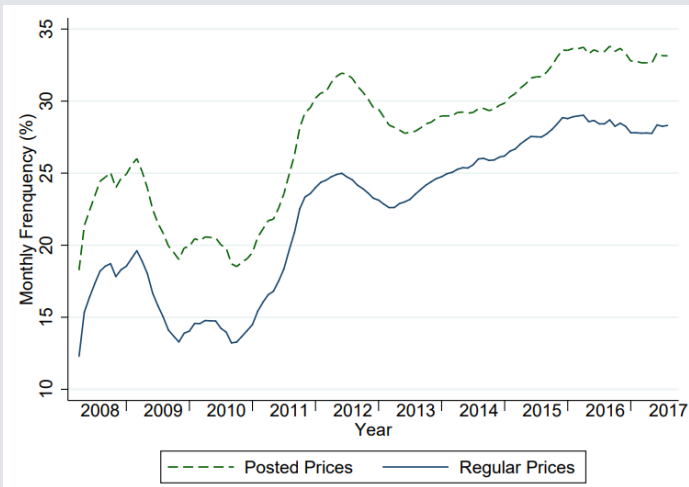
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- Theory says the FPA matters for monetary policy

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- An intentionally tractable approach: similar to a menu cost model, so we can use existing results
- Characterize how AI pricing changes the Phillips curve and monetary policy
- Provide 4 predictions for future AI pricing + monetary policy

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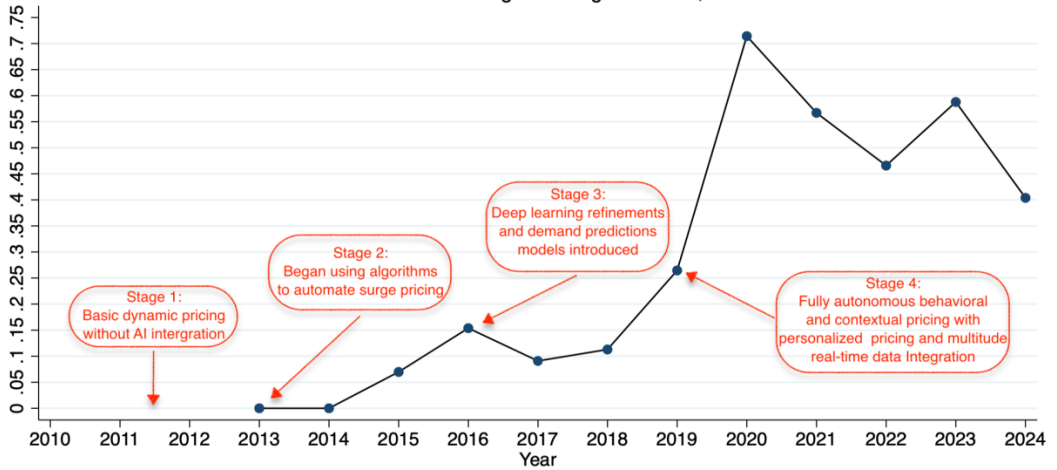
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- Model complexity is a choice
- Richer models are more costly
- But are more valuable for decision-making

What Is AI Pricing? (Example)

Uber: Fraction of AI Pricing in Pricing Job Posts, 2010-2024



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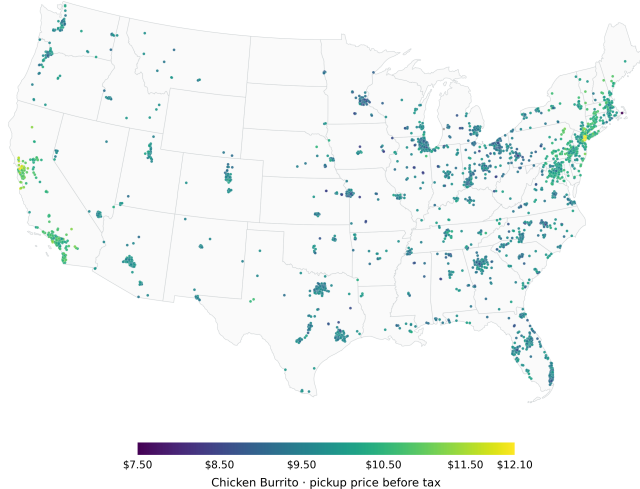
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- Firm's model complexity = how many factors $F_{ij}(k)$ to use

What is AI Pricing? (Example)

Chipotle Chicken Burrito prices

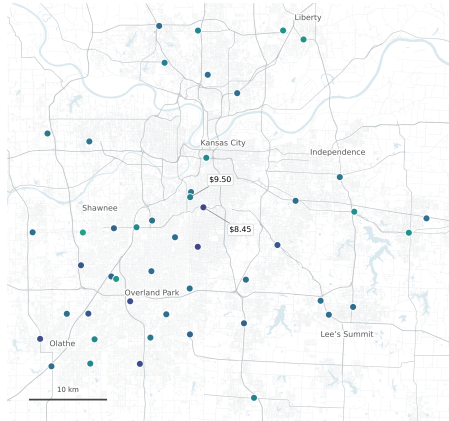


3,160 priced restaurants; 2 temporarily unavailable. Public sample, not a census. Prices observed September 8–11, 2026.

Source: FastFoodIndex · Boundaries: U.S. Census Bureau

What is AI Pricing? (Example)

Kansas City: Chipotle Chicken Burrito prices



\$7.50 \$8.50 \$9.50 \$10.50 \$11.50 \$12.10

Chicken Burrito · pickup price before tax

48 sampled stores · 12 distinct prices · September 11, 2026 snapshot
Pickup before tax · Incomplete sample · FastFoodIndex / Census TIGER/Line 2025

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 - Scales with model complexity A_i
 - $\implies$ different behavior

Gap Representation

- *Gross gap* is factor k -specific:

$$g_{it}(k) = \underbrace{p_{it}(k) - \hat{\beta}_{it}(k)}_{x_{it}(k)} - \overbrace{\omega(k)a_t}^{\text{aggregates}}$$

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- Integrating over factors k and products j , the firm's instantaneous loss is

$$\mathcal{L}_{it} = - \underbrace{\mathbf{BM} \int_0^{A_i} g_{it}(k)^2 dk}_{\text{modeled factors}} - \overbrace{\mathbf{BM}(1 - A_i)V_{\hat{z}}}^{\text{unmodeled factors}}$$

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- The inner loop decision:
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 - Looks like a menu cost model, but for a single factor k applied to M markets

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- Similar predictions for *frequency of price adjustment* (ongoing work)

An AI Phillips Curve?

- Modeled factors $k \in [0, A_i]$: Calvo-plus menu cost dynamics

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 - Special case with zero trend inflation: isomorphic to Calvo

The AI Phillips Curve

Theorem: the *Primary Eigenvalue Decomposition* implies

$$\pi_t = \Lambda(A_i) mc_t + \beta \mathbb{E}_t \pi_{t+1}$$

with slope

$$\Lambda(A_i) \equiv \Omega \left(A_i \frac{(1 - \theta_{A_i})(1 - \beta \theta_{A_i})}{\theta_{A_i}} + (1 - A_i) \frac{(1 - \theta_{\neg A_i})(1 - \beta \theta_{\neg A_i})}{\theta_{\neg A_i}} \right)$$

$$\theta_{A_i} \equiv \exp \left(-\zeta - \frac{\sigma_\beta^2 2\pi^2}{\ell^2} \right) \quad \theta_{\neg A_i} \equiv \exp(-\zeta)$$

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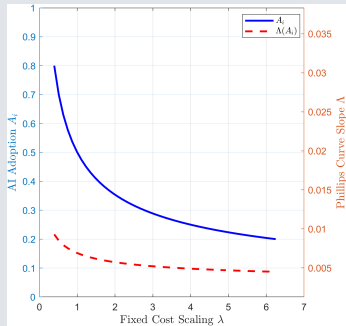
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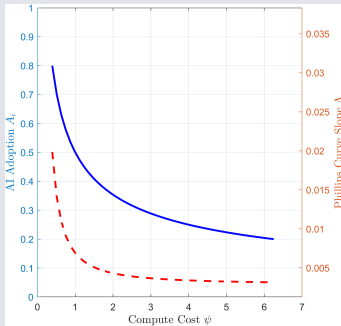
Slope $\Lambda(A_i)$ is increasing in A_i !

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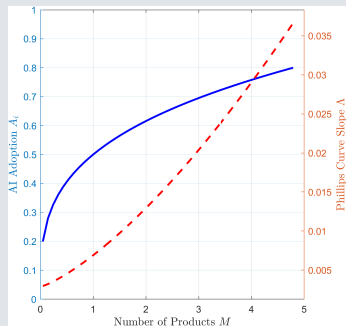
How Do AI Determinants Change the Slope?



(a) Fixed cost



(b) Compute cost



(c) Firm size

AI-NK Model

An AI-Pricing New Keynesian Model

- Replace the pricing block in a textbook New Keynesian model:

AI Phillips Curve: $\pi_t = \Lambda(A_i)\omega y_t + \beta \mathbb{E}_t \pi_{t+1} + u_t^c$

Euler equation: $\gamma y_t = \mathbb{E}_t[\gamma y_{t+1} - (i_t - \mathbb{E}_t \pi_{t+1})] + u_t^d$

Taylor rule: $i_t = \phi_\pi \pi_t + \phi_y y_t + u_t^m$

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- Explore how A_i rising affects the macroeconomy

Calibration

Object	Value	Source
Discount rate ρ	0.04	Standard annual rate
Elasticity of substitution η	6	Standard
Strategic complementarity Ω	0.55	Gagliardone et al. (2025)
Profit loss curvature $\mathbf{B}$	33.3	Implied by η, Ω
Taylor Rule ϕ_π, ϕ_y	1.5, 0.5	Standard
Baseline AI adoption $\bar{A}$	0.5	Normalization
Diffusion variance σ_β^2	0.0073	Match moments
Poisson reset rate ζ	0.85	Match moments
Inaction boundary $\bar{x}$	0.088	Match moments
FPA F	1.26	French CPI microdata (Alvarez et al., 2024)
Std. dev. of price changes	0.076	French CPI microdata
Kurtosis of price changes	5.05	French CPI microdata

What Changes with AI Adoption?

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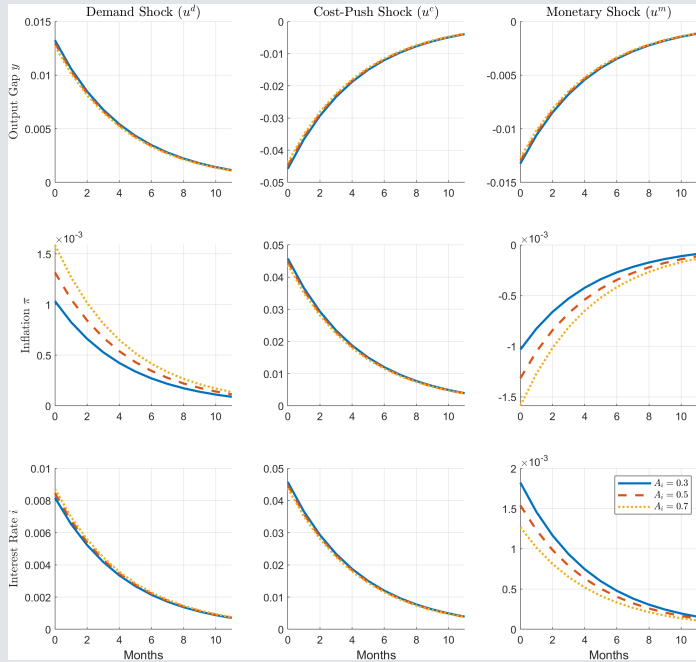
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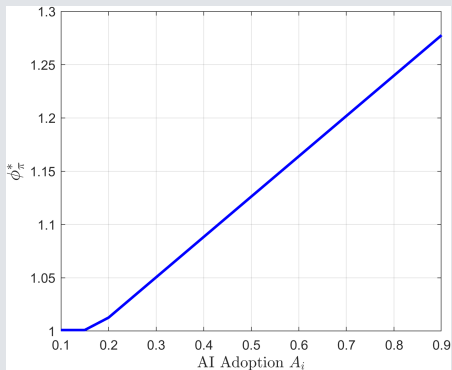
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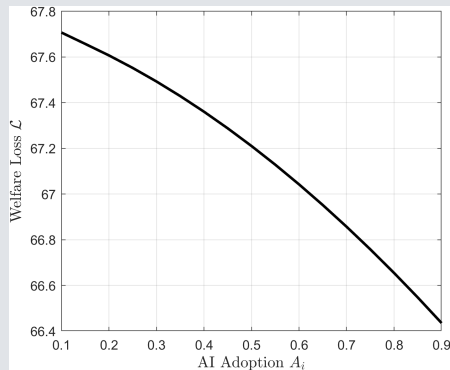
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- little change to cost-push shocks (enter the Phillips curve directly)
- Standard: changing the Phillips curve slope changes optimal monetary policy



Optimal Monetary Policy



(a) Optimal inflation response ϕ_π^*



(b) Minimized welfare loss $\mathcal{W}$

More AI pricing $\implies$ **more aggressive** monetary policy!

What's Next?

- The predictions so far:
 1. AI pricing will continue to rise as costs decline
 2. Prices will become more flexible
 3. The Phillips curve will become steeper
 4. Optimal monetary policy will respond more aggressively to inflation
- Gathering evidence relating AI use + pricing statistics
- New pricing theories!
- Really need better discipline on the AI pricing cost function

References

- Adams, Jonathan J.**, “Sticky Prices for Inflationary Economies: A Tractable Linear Approximation to Menu Cost Models with Trend Inflation,” *Federal Reserve Bank of Kansas City Working Paper*, March 2026, 26 (3).
- , **Min Fang, Zheng Liu, and Yajie Wang**, “The rise of AI pricing: Trends, driving forces, and implications for firm performance,” *Journal of Monetary Economics*, January 2026, 157, 103875.
- Alvarez, Fernando, Andrea Ferrara, Erwan Gautier, Hervé Le Bihan, and Francesco Lippi**, “Empirical Investigation of a Sufficient Statistic for Monetary Shocks,” *The Review of Economic Studies*, August 2024, p. rdae082.
- Cavallo, Alberto**, “More Amazon effects: online competition and pricing

- For each modeled factor, zero-trend-inflation dynamics are approximated by:

$$x_{k,t}^* = (1 - \beta\theta_k)a_t + \beta\theta_k\mathbb{E}_t x_{k,t+1}^*$$

$$x_{k,t} = (1 - \theta_k)x_{k,t}^* + \theta_k x_{k,t-1}$$

- Modeled and unmodeled factors differ only in effective stickiness:

$$\theta_{A_i} = \exp\left(-\zeta - \frac{\sigma_\beta^2 2\pi^2}{\ell^2}\right), \quad \theta_{\neg A_i} = \exp(-\zeta)$$

Price Change Moments

- Economy-wide price changes are a mixture:

$$F = A_i F^{CP} + (1 - A_i) F^C$$

$$\sigma_{\Delta p}^2 = \frac{A_i F^{CP} (\sigma_{\Delta p}^{CP})^2 + (1 - A_i) F^C (\sigma_{\Delta p}^C)^2}{F}$$

- Targets: frequency, standard deviation, kurtosis
- Calvo reset sizes have kurtosis 6; Calvo-plus lowers tail mass through bounded adjustments
- Baseline pins down $(\sigma_{\beta}^2, \zeta, \bar{x})$